PhenPath identifies biological functions underlying phenotypes

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Introduction

A major challenge in understanding the molecular mechanisms at the basis of diseases is the co-occurrence of different, apparently unrelated, phenotypes. Leveraging the available information on the relations among genes, diseases and phenotypes, we developed PhenPath[1], a resource dedicated to collect and enrich knowledge on molecular functions and biological processes underlying the occurrence of phenotypes.

PhenPath associates **phenotypes** to **pathways** exploiting relationships among genes, diseases & phenotypes

Associations between diseases and phenotypes derive from HPO and OMIM

Given a set of phenotypes, it retrieves the **shared biological pathways**, genes and diseases

7,137 phenotypes (the 47% of HPO) associated with **6,838** pathways

PhenPath methods and results

PhenPath associates phenotypes with diseases and genes via eDGAR[2], and it retrieves functional annotation for each phenotype after an enrichment procedure using NET-GE[3]. Network-based enrichment analysis identifies significantly over-represented biological functions and pathways, considering Gene Ontology (Molecular Function, Biological Process and Cellular Component), KEGG and Reactome pathways, as reported in the following table:

	# OF PHENOTYPES	# OF ASSOCIATED ENTRIES
PhenPathDB (HPO IDs)	12,111	
with associated diseases	7,137 (58.9%)	4,292 diseases 3,446 genes
with NET-GE enriched BP GO terms	6,256 (51.7%)	6,838 BP GO terms
with NET-GE enriched MF GO terms	6,202 (51.2%)	2,211 MF GO terms
with NET-GE enriched CC GO terms	5,254 (43.4%)	946 CC GO terms
with NET-GE enriched KEGG	4,198 (34.6%)	326 KEGG
with NET-GE enriched Reactome	5,550 (45.8%)	1,369 REACTOME

PhenPathTOOL: a case study

Based on PhenPath, we developed a tool that associates a set of phenotypes (possibly co-occurring in a disease) with potentially involved genes and new functional terms. These result derive from the intersection of the sets of genes and functional terms linked to each phenotype in PhenPath. We show here the case study of the co-occurrence Obesity, Ovarian Cyst and Diabetes. PhenPathTOOL allows to retrieve the diseases and at the genes shared among the input phenotypes. Moreover, based on the enrichment analysis performed with NET-GE [3], PhenPathTOOL reports the molecular functions and the biological pathways shared among the phenotypes, highlighting possible explanation for their co-occurrence in the same patient.

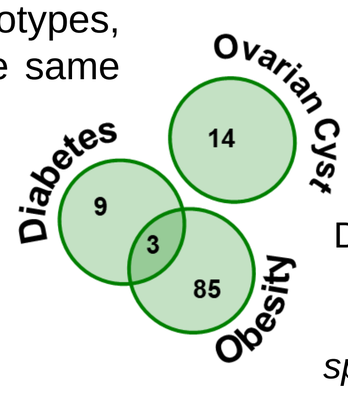
PhenPath could be used to study a complex disease, as we proved in many study cases reported in the publication (e.g.: Rett syndrome, Tourette syndrome).

Ovarian Cyst (HP:0000138)

Obesity (HP:0001513)

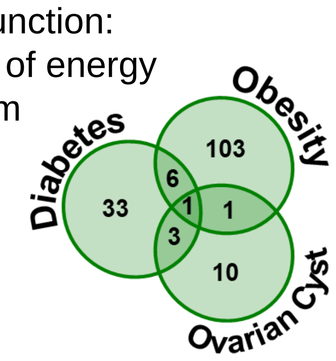
Diabetes (HP:0005978)

PhenPathTOOL



Shared diseases
Diabetes and Obesity share 3 diseases: *Morbid obesity spermatogenic failure, Microcephalic osteodysplastic Primordial dwarfism, type II*

Shared genes
The 3 phenotypes share PPARG, the *peroxisome proliferator-activated receptors gamma*; function: regulation of energy metabolism



NET-GE performs the network-based enrichment of GO terms and KEGG/Reactome pathways

Comparison of annotation retrieved for each phenotype

KEGG pathways
The 3 phenotypes share 3 terms
eg: *Signal transduction*

Reactome pathways
10 terms are shared among pairs of phenotypes, eg: *Signaling by Insulin-like Growth Factor 1 Receptor (IGF1R)*

Biological Processes (GO)
The 3 phenotypes share 58 terms
eg: *Response to insulin*

Molecular Function (GO)
The 3 phenotypes share 7 terms
eg: *Hormone receptor binding*

Cellular Component (GO)
The 3 phenotypes share 4 terms
eg: *Organelle membrane*

Conclusions

We propose our resource for formulating hypothesis on the relations among phenotypes and underlying molecular mechanisms. We participated to the Critical Assessment of protein Function Annotation algorithms (CAFA) 4th edition. Preliminary results of the experiment show that PhenPath is among the top-scoring methods for predicting the association between genes and phenotypes.

References:
[1] Babbi G, Martelli PL, Casadio R. PhenPath: a tool for characterizing biological functions underlying different phenotypes BMC Genomics. 2019
[2] Babbi G, Martelli PL, Profiti G, Bovo S, Savojardo C, Casadio R. *eDGAR: a database of Disease-Gene Associations with annotated Relationships among genes*. BMC Genomics. 2017
[3] Bovo S, Di Lena P, Martelli PL, Fariselli P, Casadio R. *NET-GE: a web-server for NETwork-based human Gene Enrichment*. Bioinformatics. 2016